

AI STUDY PARTNER

Complete A to Z Setup Guide

Telegram Bot • Gemini AI • Google Sheets • Free 24/7 Hosting on Render

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What You Are Setting Up

AI Study Partner is an autonomous study-mentor Telegram bot. A student just chats with it; the bot designs a study plan, tracks progress, runs quizzes, generates slides and research, and manages everything inside an auto-created Google Sheet. This guide takes you from zero to a live, always-on bot.

The big picture — 5 things to wire together

- **Telegram** – the chat interface students use (bot token).
- **Gemini AI** – the brain that plans, quizzes and explains (API key).
- **Google Sheets** – the bot's memory / planner (service-account JSON).
- **Render** – free cloud host that runs the bot 24/7 (web service).
- **UptimeRobot** – free pinger that guarantees it never sleeps.

TIP Total cost is **\$0**. Every service here has a free tier that is more than enough for a workshop or a classroom.

Accounts you will need

Account	Used for	Cost
Telegram	Create the bot, get the token	Free
Google AI Studio	Gemini API key	Free
Google Cloud	Sheets + Drive service account	Free
GitHub	Hold the code Render deploys from	Free
Render	Host the bot 24/7	Free
UptimeRobot	Keep it awake (recommended)	Free
Supabase	Persistent database (optional)	Free

IMPORTANT Set aside about **30-40 minutes** the first time. Most of it is creating free accounts and copying keys. The bot needs no coding from you.

Part 1 — Get Your Keys & Credentials

STEP 1 Create the Telegram bot

- Open Telegram, search for **@BotFather**, open the chat.
- Send the command **/newbot**.
- Choose a display name, e.g. *AI Study Partner*.
- Choose a username ending in **_bot**, e.g. *aistudypartner_bot*.
- BotFather replies with a **token** – copy it.

```
TELEGRAM_BOT_TOKEN = 8123456789:AAH...your-long-token...xyz
```

TIP Keep this token secret. Anyone who has it can control your bot.

STEP 2 Get your free Gemini API key

- Go to <https://aistudio.google.com/apikey>
- Sign in with any Google account.
- Click **Create API Key**, then copy the key.

```
GEMINI_API_KEY = AIzaSy...your-key...
```

STEP 3 Create the Google Sheets service account

This lets the bot create and edit Google Sheets on its own.

- Go to <https://console.cloud.google.com> and create a new project (name it *AI Study Partner*).
- In the search bar, open **Google Sheets API** and click **Enable**.
- Search **Google Drive API** and click **Enable**.
- Open **APIs & Services > Credentials**.
- Click **Create Credentials > Service Account**. Name it *study-partner-bot*, then Create and Done.
- Click the new service account, open the **Keys** tab, then **Add Key > Create new key > JSON**.
- A **credentials.json** file downloads. Keep it safe.

IMPORTANT Inside **credentials.json** there is a field **client_email** (it looks like *...@...iam.gserviceaccount.com*). Copy it – that is your **SHEET_SHARE_EMAIL**.

```
SHEET_SHARE_EMAIL = study-partner-bot@your-project.iam.gserviceaccount.com
```

STEP 4 Convert credentials.json to one line (base64)

Render stores secrets as text, so we encode the JSON file into a single base64 string for the **GOOGLE_CREDENTIALS_JSON** variable.

On Mac / Linux

```
base64 -i credentials.json | tr -d '\n'
```

On Windows (PowerShell)

```
[Convert]::ToBase64String([IO.File]::ReadAllBytes("credentials.json"))
```

Copy the long output – that single line is your **GOOGLE_CREDENTIALS_JSON** value.

TIP Running locally instead? You can skip base64 – just keep credentials.json in the project folder and set **GOOGLE_CREDENTIALS_JSON=./credentials.json**

Part 2 — Get the Code onto GitHub

Render deploys from a GitHub repository. The project lives in the **ai_study_partner/** folder of your repo.

- If the code is already in your GitHub repo, you are done with this part – skip to Part 3.
- To work on it locally first, clone it:

```
git clone https://github.com/justmything04-lang/CHAT.git
cd CHAT/ai_study_partner
```

TIP The whole project is self-contained in the **ai_study_partner/** folder, so Render only needs that folder as its root directory.

Part 3 — Deploy to Render (Free Web Service)

STEP 5 Create the web service

- Go to <https://render.com> and sign in with GitHub.
- Click **New > Web Service**.
- Select your repository (*CHAT*).

STEP 6 Configure the service

Setting	Value
Root Directory	ai_study_partner
Language / Env	Python 3
Build Command	pip install -r requirements.txt
Start Command	python main.py
Instance Type	Free
Health Check Path	/health

IMPORTANT The **Root Directory** must be **ai_study_partner**, otherwise Render will not find `requirements.txt` or `main.py`.

STEP 7 Add the environment variables

In the Render **Environment** tab, add these keys (values from Part 1):

Variable	Value / Note	Required?
TELEGRAM_BOT_TOKEN	from BotFather	Yes
GEMINI_API_KEY	from AI Studio	Yes
GOOGLE_CREDENTIALS_JSON	the base64 line	Yes
SHEET_SHARE_EMAIL	service-account email	Yes
STORAGE_BACKEND	sqlite (or supabase)	Yes
TIMEZONE	Asia/Kolkata	Yes
KEEP_ALIVE_SECONDS	600	Yes
RENDER_APP_URL	added in Step 9	Later

STEP 8 Deploy and grab your URL

- Click **Create Web Service**. Render installs and starts the bot.
- Wait for the log line **Bot starting in long-polling mode**.
- At the top of the page copy your service URL, e.g. <https://ai-study-partner.onrender.com>

STEP 9 Turn on the 24/7 self-ping

- Add one more environment variable:
- **RENDER_APP_URL** = the URL you just copied.
- Save. Render redeploys automatically.

```
RENDER_APP_URL = https://ai-study-partner.onrender.com
```

Now the bot pings itself every 10 minutes so the free instance never sleeps.

Part 4 — Make It Truly Always-On

STEP 10 Verify the health endpoint

Open this URL in your browser:

```
https://your-app.onrender.com/health
```

You should see: `{"status": "ok", "service": "ai-study-partner"}`

STEP 11 Add an external monitor (recommended)

The built-in self-ping keeps the bot awake while it is running, but it cannot wake a process that has already slept. A free external monitor closes that gap.

- Go to <https://uptimerobot.com> and sign up free.
- Click **Add New Monitor > HTTP(s)**.
- URL: `https://your-app.onrender.com/health`
- Monitoring interval: **5 minutes**, then Create.

TIP Self-ping plus UptimeRobot together give genuine round-the-clock uptime on Render's free plan.

IMPORTANT With `STORAGE_BACKEND=sqlite`, student data resets whenever Render redeploys (its disk is temporary). For a one-day workshop that is fine. To keep data permanently, use `STORAGE_BACKEND=supabase` (see Part 6).

Part 5 — Test & Use the Bot

STEP 12 Talk to your bot

- Open Telegram and search your bot's `@username`.
- Send `/start`. It should ask for your name.
- Give a goal like *CA Inter GST in 20 days* and a target date.
- It builds a plan and shares a Google Sheet link. You are live!

Command cheat-sheet

Command	What it does
<code>/start</code>	Onboard and create the study plan + sheet
<code>/today</code>	Today's study targets
<code>/done [topic]</code>	Mark a topic complete

Command	What it does
/test [topic]	5-question MCQ quiz
/stuck [topic]	Flag a weak area
/progress	Progress report + dashboard snapshot
/reschedule	Reshuffle the remaining plan
/research [topic]	Deep research summary
/explain [topic]	Plain-language explanation
/slides [topic]	Generate a Gamma slide-deck link
/upload [subject]	Upload a PDF (NotebookLM)
/ask [question]	Ask from your uploaded PDF
/settings	Change briefing time and alert frequency
/help	List every command

TIP Students can also just chat normally – "I finished chapter 3" works the same as /done.

Part 6 — Optional Upgrades

A. Persistent database with Supabase

- Create a free project at <https://supabase.com>
- Open **SQL Editor**, paste the contents of **supabase_setup.sql**, Run.
- From **Project Settings > API**, copy the URL and the key.
- On Render set: `STORAGE_BACKEND=supabase`, `SUPABASE_URL=...`, `SUPABASE_KEY=...`

B. Slides with Gamma (no API key)

- Create a free account at <https://gamma.app>
- Stay logged in there in your browser.
- When a student uses `/slides`, the bot sends a one-click link that opens a pre-filled deck – they just hit Generate.

C. Study from your own PDFs (NotebookLM)

Optional and unofficial. If installed, `/upload`, `/ask` and `/podcast` use NotebookLM; if not, the bot automatically falls back to Gemini.

```
pip install git+https://github.com/teng-lin/notebooklm-py.git
```

Troubleshooting

Symptom	Fix
Bot does not reply at all	Check Render logs. Confirm <code>TELEGRAM_BOT_TOKEN</code> is correct.
409 Conflict in logs	Another instance/webhook is active. Redeploy; the bot auto-clears webhooks.
Sheet not created / permission error	Enable Sheets API + Drive API; recheck <code>GOOGLE_CREDENTIALS_JSON</code> base64.
Gemini errors / empty replies	Verify <code>GEMINI_API_KEY</code> and that you have free quota left.
Bot sleeps after 15 minutes	Set <code>RENDER_APP_URL</code> and add the UptimeRobot monitor.
Data lost after a redeploy	Expected on sqlite. Switch <code>STORAGE_BACKEND</code> to supabase.
Briefings not arriving	Service must stay awake (Part 4). Check <code>TIMEZONE</code> is correct.

Environment Variables — Full Reference

Variable	Required	Example / Default
<code>TELEGRAM_BOT_TOKEN</code>	Yes	8123...:AAH...

Variable	Required	Example / Default
GEMINI_API_KEY	Yes	AlzaSy...
GOOGLE_CREDENTIALS_JSON	Yes	base64 string
SHEET_SHARE_EMAIL	Yes	...@...iam.gserviceaccount.com
STORAGE_BACKEND	Yes	sqlite supabase gsheets
TIMEZONE	Yes	Asia/Kolkata
RENDER_APP_URL	Yes (host)	https://app.onrender.com
KEEP_ALIVE_SECONDS	No	600
SUPABASE_URL	If supabase	https://xxx.supabase.co
SUPABASE_KEY	If supabase	eyJ...
GAMMA_BASE_URL	No	https://gamma.app/create
GAMMA_DEFAULT_CARDS	No	10

You now have a fully autonomous, always-on AI Study Partner. Happy teaching!